



TEAM 5 – Dashboard, Integration & Automation

ThreatGraph Project Documentation Guide

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Organization: CyArt Tech

Domain: Security Operations Center (SOC)

Project Type: Cybersecurity / CTEM / Attack Surface Management



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1. Introduction

Modern organizations are continuously exposed to cyber threats due to the increasing number of internet-facing assets, cloud infrastructure, APIs, remote access services, and third-party integrations. Traditional security approaches often rely on periodic vulnerability assessments and manual monitoring, which are insufficient for detecting rapidly evolving cyber threats.

The ThreatGraph project is designed to provide continuous visibility into an organization's attack surface by integrating asset discovery, monitoring, vulnerability analysis, attack path visualization, and centralized dashboarding.

Team 5 focuses on Dashboard, Integration, and Automation, which acts as the centralized operational layer of the ThreatGraph platform. The team is responsible for collecting data from different modules, integrating Security Information and Event Management (SIEM) systems, visualizing security insights through dashboards, automating alerts and workflows, and enabling centralized threat monitoring.

The proposed solution primarily leverages open-source technologies such as Elasticsearch, Kibana, Logstash, MISP, and Python automation frameworks to build a scalable and cost-effective cybersecurity monitoring environment.

2. Problem Statement

Most organizations face several cybersecurity monitoring challenges such as:

- Lack of centralized visibility across infrastructure
- Distributed logs and fragmented monitoring systems
- Delayed incident detection and response
- Manual threat analysis processes
- High alert fatigue caused by false positives
- Difficulty correlating data from multiple security tools
- Limited automation in incident workflows

Traditional systems generally rely on isolated tools that are unable to provide a unified security overview. As organizations scale, managing logs, vulnerabilities, threat intelligence, and incidents becomes increasingly complex.

The ThreatGraph Dashboard & Integration module aims to solve these challenges by building a centralized monitoring platform capable of:

- Real-time log analysis
- Centralized dashboard visualization
- Threat intelligence enrichment
- Automated alert generation
- Ticketing integration
- Continuous monitoring and reporting



3. Objectives

The major objectives of Team 5 are:

- To design a centralized cybersecurity dashboard
- To integrate SIEM-based monitoring into ThreatGraph
- To automate alerting and incident workflows
- To integrate external threat intelligence feeds
- To provide real-time visualization of security events
- To improve incident response efficiency
- To support scalable monitoring architecture using open-source tools
- To reduce manual monitoring efforts through automation

4. Scope of the Project

The scope of Team 5 includes:

Dashboard Development

- Security monitoring dashboard
- Risk visualization
- Asset monitoring visualization
- Attack path visualization integration

SIEM Integration

- Log aggregation
- Event correlation
- Security alerting
- Centralized monitoring

Threat Intelligence Integration

- IOC enrichment
- Reputation analysis
- Threat feed correlation

Automation

- Automated ticket creation
- Alert forwarding
- Incident notification
- Report generation

Reporting

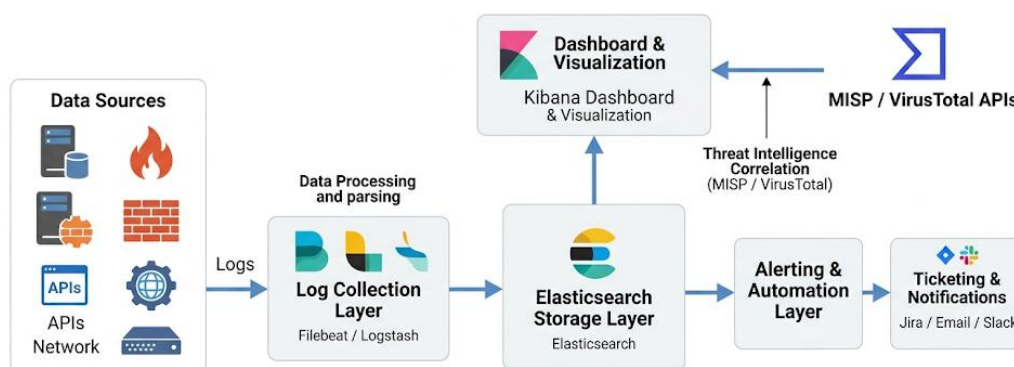
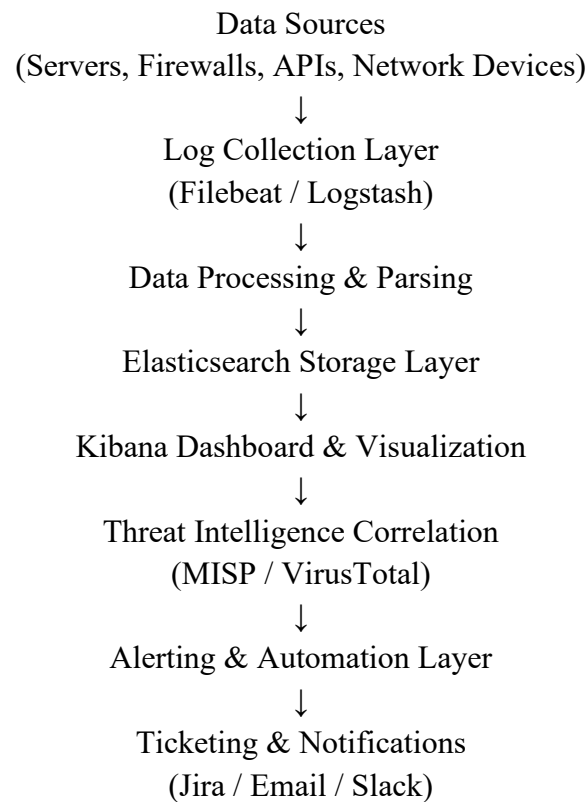
- Security reports
- Monitoring statistics
- Incident summaries



- Risk trend analysis

5. Proposed Architecture

High-Level Architecture Flow



Team 5 - High-Level Architecture Flow: Dashboard, Integration & Automation

Architecture Components



1. Data Sources

Security logs are collected from:

- Linux servers
- Windows systems
- Firewalls
- Routers
- APIs
- Applications
- Authentication services

2. Log Collection Layer

Filebeat and Logstash are used for collecting and forwarding logs into Elasticsearch.

3. Storage Layer

Elasticsearch acts as the centralized storage and indexing engine.

4. Dashboard Layer

Kibana is used for creating visual dashboards and monitoring panels.

5. Threat Intelligence Layer

MISP and VirusTotal APIs are integrated for IOC enrichment.

6. Automation Layer

Python scripts and automation workflows trigger alerts, notifications, and ticket creation.

6. Workflow & Execution Process

Step 1 – Collecting Logs

Logs are collected from multiple systems using:

- Filebeat
- Logstash
- Syslog forwarding

Example Sources

- Authentication logs
- Firewall logs
- Network device logs
- Web server logs
- Application logs

Step 2 – Parsing & Normalization

Logstash is configured to:



- Parse incoming logs
- Normalize fields
- Remove unnecessary data
- Convert logs into structured JSON format

Step 3 – Elasticsearch Indexing

Parsed logs are indexed into Elasticsearch for:

- Fast searching
- Data aggregation
- Analytics
- Correlation

Step 4 – Dashboard Visualization

Kibana dashboards display:

- Failed login attempts
- High-risk alerts
- Top source IPs
- IOC matches
- Traffic trends
- Security incidents

Step 5 – Threat Intelligence Correlation

Threat intelligence feeds from:

- MISP
- VirusTotal
- Open Threat Exchange (OTX)

are used to enrich suspicious IPs, domains, and hashes.

Step 6 – Automation & Alerting

Automation workflows:

- Generate alerts
- Send email notifications
- Create Jira tickets
- Trigger webhook actions

Step 7 – Reporting

Periodic reports are generated containing:

- Incident summaries
- Risk trends
- IOC detections
- Monitoring statistics



7. Open-Source Tools & Technology Stack

Tool	Purpose
Elasticsearch	Data indexing & search
Kibana	Dashboard visualization
Logstash	Log parsing & processing
Filebeat	Lightweight log collection
MISP	Threat intelligence platform
VirusTotal API	IOC enrichment
Python	Automation scripting
Jira	Ticket management
Ubuntu Linux	Deployment environment
Docker	Containerized deployment

8. Tool Justification

1.Elasticsearch

Elasticsearch was selected because it provides:

- High-speed indexing
- Scalable distributed architecture
- Powerful search capabilities
- Strong community support
- Easy integration with Kibana

Compared to traditional relational databases, Elasticsearch performs significantly better for large-scale log analytics and real-time querying.

2.Kibana

Kibana was chosen because:

- It provides real-time dashboard visualization
- It integrates directly with Elasticsearch
- It supports customizable charts and graphs
- It simplifies SOC monitoring workflows

3.Logstash

Logstash was selected for:

- Flexible log parsing
- Support for multiple input formats
- Data transformation capabilities
- Open-source extensibility



4.Filebeat

Filebeat provides:

- Lightweight log forwarding
- Low system overhead
- Easy deployment
- Reliable event shipping

5.MISP

MISP was selected because:

- It is open-source
- Supports IOC sharing
- Provides threat intelligence correlation
- Has strong community adoption

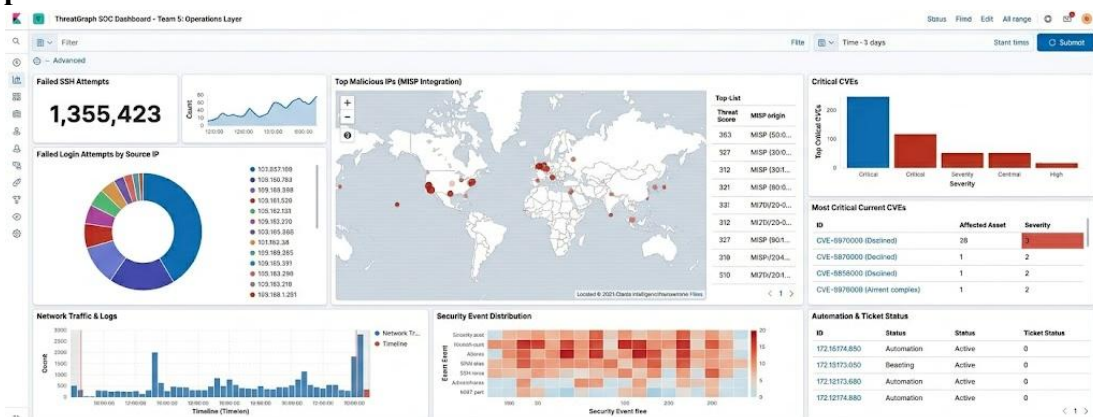
6.Python

Python is used because:

- Easy automation scripting
- Large cybersecurity ecosystem
- Extensive API support
- Easy integration with SIEM workflows

9. Dashboard Design & Visualization

Proposed Dashboard Panels



ThreatGraph SOC Dashboard - Team 5: Operations Layer

Security Overview Panel

Displays:

- Total assets monitored
- Active alerts
- Critical vulnerabilities



- Risk score summary

Authentication Monitoring Panel

Displays:

- Failed login attempts
- SSH brute-force attempts
- Suspicious login locations

Threat Intelligence Panel

Displays:

- Malicious IP detections
- IOC matches
- VirusTotal reputation results

Vulnerability Monitoring Panel

Displays:

- CVE statistics
- Risk prioritization
- Severity distribution

Network Monitoring Panel

Displays:

- Traffic spikes
- Top source IPs
- Protocol distribution

Visualization Types

- Pie charts
- Heatmaps
- Timeline graphs
- Bar charts
- Line graphs
- Data tables

10. SIEM Integration

The SIEM integration layer acts as the centralized monitoring component.

Major Functions

- Event aggregation
- Log correlation
- Threat detection
- Alert generation



- Incident monitoring

Data Flow

System Logs → Logstash → Elasticsearch → Kibana → Alerts

Security Events Monitored

- Failed logins
- Unauthorized access attempts
- Suspicious IP activity
- Malware indicators
- Firewall alerts

11. Threat Intelligence Integration

Threat intelligence enhances the accuracy of detection by enriching logs with external threat data.

Integrated Sources

- MISP
- VirusTotal
- AlienVault OTX

IOC Types

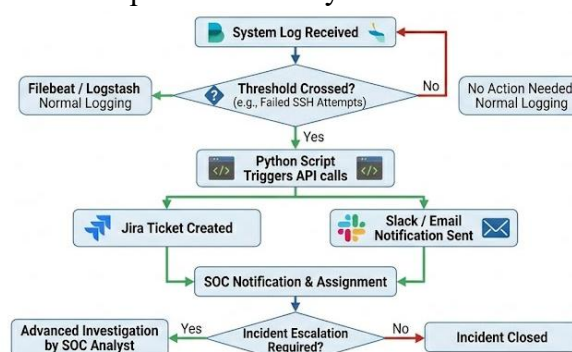
- Malicious IP addresses
- Domains
- URLs
- File hashes

Workflow

Suspicious IOC → Threat Feed Lookup → Reputation Analysis → Alert Generation

12. Automation Workflow

Automation improves incident response efficiency and reduces manual workload.



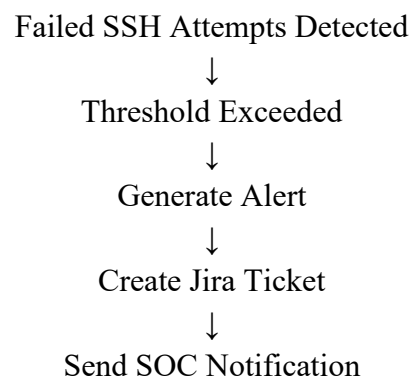
ThreatGraph Automation & Incident Response Workflow (Team 5 - Operations Layer)



Automation Features

- Automatic alert creation
- Ticket generation
- Email notifications
- Slack integration
- Incident escalation

Example Workflow



Python Automation Usage

Python scripts can:

- Fetch IOC data
- Trigger APIs
- Parse logs
- Send alerts
- Generate reports

13. Security & Scalability Considerations

Security Measures

- Role-Based Access Control (RBAC)
- HTTPS encryption
- Secure API authentication
- Log integrity validation
- Secure dashboard access

Scalability Considerations

- Distributed Elasticsearch cluster
- Horizontal scaling
- Docker container deployment
- Cloud compatibility
- Load balancing

Performance Optimization



- Index lifecycle management
- Data retention policies
- Query optimization
- Log filtering

14. Challenges & Solutions

Challenge	Solution
Large log volume	Log filtering & indexing optimization
False positives	Alert tuning & IOC correlation
Integration complexity	API-based modular integration
Dashboard performance	Efficient query optimization
Alert fatigue	Risk-based prioritization

15. Optimization Strategies

Techniques Used

- Log normalization
- Alert deduplication
- IOC enrichment
- Dashboard optimization
- Automated filtering
- Data retention management

Benefits

- Faster detection
- Reduced false positives
- Better SOC efficiency
- Improved incident response

16. Future Enhancements

Future improvements may include:

- AI-based alert prioritization
- Machine learning anomaly detection
- SOAR integration
- Cloud-native monitoring
- Advanced attack path correlation
- Multi-tenant dashboards
- Automated threat remediation



17. Conclusion

The Dashboard, Integration & Automation module serves as the centralized operational layer of the ThreatGraph platform. By integrating SIEM systems, threat intelligence feeds, and automation workflows, the proposed architecture provides continuous visibility into organizational security posture.

The solution focuses on scalability, open-source technologies, centralized monitoring, and automation-driven incident response. The integration of Elasticsearch, Kibana, MISP, and Python-based automation creates an efficient and practical SOC-oriented monitoring environment.

This module significantly improves threat visibility, incident response efficiency, and security operations management while maintaining flexibility and scalability for future enhancements.

18. References

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